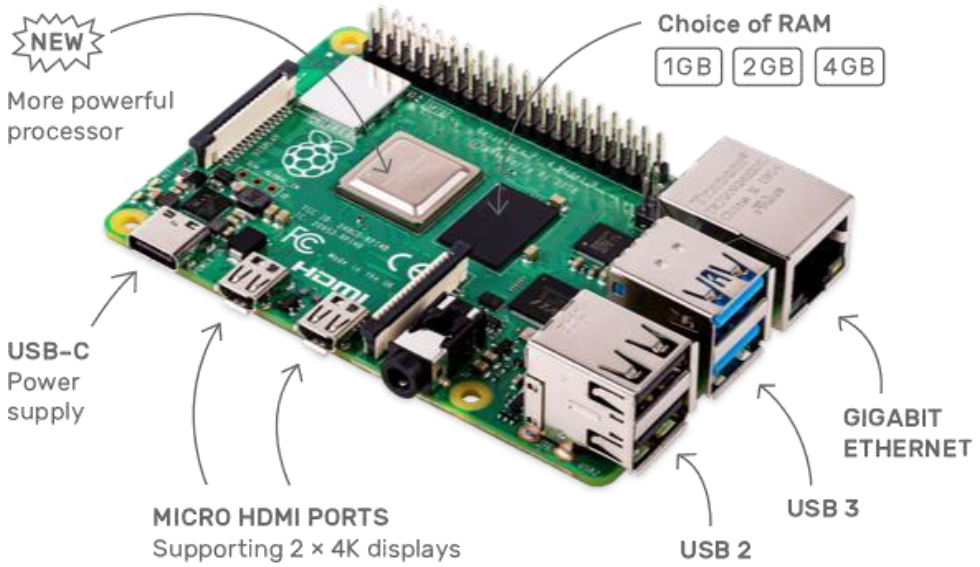
EECS 388 Lab #8 – Option 1

Board-to-Board Communication

Choose this option if you want to continue learning about Raspberry PIs and use the HiFive board in your Final Lab project.

In this lab, you will establish UART based communication channels between the Pi 4 and the HiFive1 board.



Part 1: Setup the UART connections

In this part, we will connect the HiFive1 and the Raspberry Pi 4 boards via two UART channels. While the Pi 4 has 4 UARTs, we will only be using two of them today (uart2 and uart3). Examine the file `/boot/config.txt` for the following two lines enabling uart2 and uart3, add them to the file if they are not present**.**

If the following two lines are already present in the config.txt file, then don't need to write or edit anything in the file.

```
dtoverlay=uart2,115200
dtoverlay=uart3,115200
```

If you needed to add these two lines, reboot the Raspberry Pi. After rebooting the system, `/dev/ttyAMA1` and `/dev/ttyAMA2` will be created. (**Again, if the lines are already in the file, don't do anything**)

Connect HiFive's UART1 RX (pin7) to Raspberry Pi 4's UART2 TX (pin 27). This is the main communication line between the Pi 4 and the HiFive1. From the Pi 4, you can access the channel via `/dev/ttyAMA1`.

For debugging of HiFive 1, connect HiFive1's UART0 TX (pin1) to Pi 4's UART3 RX (pin 29). From the Pi 4, it can be accessed via /dev/ttyAMA2.

In summary, you will be able to access the following two files from the Pi 4.

```
/dev/ttyAMA1      Pi 4 → HiFive1: Send steering angle to HiFive1 (uart1).
/dev/ttyAMA2     HiFive1 → Pi 4: Receive HiFive1's console (uart0) output
```

Part 2: Programming the HiFive1

In this part of the lab, you will program the HiFive1 to receive data from the Pi 4.

On your PC (not Pi 4), download the starter code.

Sign in to the Canvas account and go to the EEC 388 Embedded Systems section. Inside the "Files" section, go to the "Source Codes" folder. Click on the **Lab09_option1.zip** file to download the source code for lab9. Extract the file inside a folder of your PC. After that, add this folder to the VS Code workspace (like the previous lab)

Your task is to receive the data from HiFive1's UART1 channel and send the received data to UART0 channel **as a null-terminated string**. The following is rough **pseudo code** for the task, but there are other ways to accomplish the task as well.

```
while (1) {

  if (is UART1 ready?)  {
    data = read from UART1.
    print data to UART0.
  }
}
```

To implement the task, you should use the provided serial API shown in the following. Note that device id ‘`devid`’ is 0 to access UART0, while it is 1 to access UART1. For this lab, the two functions `ser_printline` and `ser_readline` will be especially helpful (since you need to send the data as a string).

```
void ser_setup(int devid);
int  ser_isready(int devid);
void ser_write(int devid, char c);
void ser_printline(int devid, char *str);
char ser_read(int devid);
int  ser_readline(int devid, int n, char *str);
```

In particular, you may need to use `ser_isready()` function to check whether a given UART channel has pending data to read. To better understand what a specific function is doing, you can always check both `eecs388_lib.h` and `eecs388_lib.c` files.

```
int  ser_isready(int devid)
{
  uint32_t regval = *(volatile uint32_t *)(UART_ADDR(devid) + UART_IP);
  return regval;
}
```

How to test your HiFive code: Once you finish programming the HiFive1, **switch to the Raspberry Pi 4** and open two terminals: one for sending data to the HiFive1, and one for seeing the debug message output from the HiFive1.

Sender’s terminal (terminal 1):

```
$ screen /dev/ttyAMA1 115200
```

Debug terminal (terminal 2):

```
$ screen /dev/ttyAMA2 115200
```

Now, type any string on the ‘terminal 1’. If you programmed your HiFive 1 correct, you should see the message coming out from the ‘terminal 2’ terminal.

Hints:

- If you see error message like "Screen is terminating", you should kill the terminals and then again re-open the terminal. You can also try running “sudo screen /dev/ttyAMA1 115200”.
- You may need to restart your HiFive board (remove and re-plug the USB power cable from the PC side so you don’t damage the HiFive boards!) depending on how large the string buffer you created if you stop seeing your output on the Raspberry Pi terminal.
- Double check all connections if you aren’t seeing your output on the Raspberry Pi. Also, the Pi may create other devices than ttyAMA1 or ttyAMA2, check which devices are created in the /dev/ folder.

Submission – 30 points (Max 31.5/30 with bonus)

Attendance: To obtain the 25% (7.5 points), you must attend the entire lab period and work toward the demo. If you are more than 30 minutes late, it will be counted as absent.

Demo: To obtain the 50% (15 points) for this lab, you need to complete the HiFive1’s code and show the complete message transfer from the Raspberry Pi4 to the Hive1 back to the second terminal of the Raspberry Pi4.

Possible questions that may be asked during the lab demo by your GTA:

1. How does the UART_ADDR() macro work for selecting the correct UART on the HiFive1?
2. What is the purpose of ser_read() and ser_write() function in the code?
3. If you want to send character data instead of string data (or vice versa) from Pi4 to HiFive1, what would be the difference in the code?
4. Where in the ser_setup() code does it enable UART interrupts?
5. Why is the ser_isready() function used in the code?
6. Does this lab handle the UART with an interrupt service routine or by polling?

Code: To obtain the 25% (7.5 points) go to the “Assignments” section on Canvas and submit your modified eecs388_comm.c file to the Lab 8 submission link. Make sure you have demonstrated your lab to your GTA before you submit your code. Screenshots/PDF/Text files will not be accepted. You need to submit the modified eecs388_comm.c file only.

Bonus: To obtain the 5% (1.5 points) you need to submit the file to Canvas by the end of your lab period. You must demo your lab to your GTA before submitting.

Appendix

GPIO mapping of Pi 4.

Function	Pin Number	Pin Number	Function
3V3	1	2	5V
SPI3 MOSI/SDA3	3	4	5V
SPI3 SCLK/SCL3	5	6	GND
SPI4 CE0 N/SDA 3	7	8	TXD1/SPI5 MOSI
GND	9	10	RXD1/SPI5 SCLK
	11	12	SPI6 CEO N
SPI6 CE1 N	13	14	GND
SDA6	15	16	SCL6
3V3	17	18	SPI3 CE1 N
SDA5	19	20	GND
RXD4/SCL4	21	22	SPI4 CE1 N
SCL5	23	24	SDA4/TXD4
GND	25	26	SCL4/SPI4 SCLK
SPI3 CE0 N/TXD2/SDA6	27	28	SPI3 MISO/SCL6/RXD2
SPI4 MISO/RXD3/SCL3	29	30	GND
SPI4 MOSI/SDA4	31	32	SDA5/SPI5 CEO N/TXD5
SPI5 MISO/RXD5/SCL5	33	34	GND
SPI6 MISO	35	36	SPI1 CE2 N
SPI5 CE1 N	37	38	SPI6 MOSI
GND	39	40	SPI6 SCLK
I2C			Ground
UART			5V Power
SPI			3V3 Power

Source: https://learn.pi-supply.com/make/raspberry-pi-4-pinout